BIS 629 Lecture 3: Design and Analysis of Cluster Randomized Trials in Implementation and Prevention Science - Part 2

Donna Spiegelman

Center for Methods in Implementation and Prevention Science (CMIPS)
Department of Biostatistics

Yale School of Public Health

Derivation of the non-centrality parameter – linear mixed models

We derived in Slide 3.27 that

$$Var(\hat{\beta}^*) = Var(\bar{Y}_1 - \bar{Y}_2) = 2\frac{Var(Y_{ijk})}{Km}IF = 2\frac{\sigma^2}{Km}[1 + (m-1)\rho]. \quad (1)$$

Recall that the non-centrality parameter for the Wald test is $\lambda = \frac{(\beta^*)^2}{Var(\hat{\beta}^*)}$. Now, we substitute the variance for the difference between two means for independent data with a revised variance for a continuous outcome in a CRT, and obtain $\lambda = \frac{(\beta^*)^2}{2\frac{\sigma^2}{Km}[1+(m-1)\rho]}$, where here $\beta^* = \mu_1 - \mu_2$. That is, we multiply the denominator of the non-centrality parameter for independent (not clustered) data by the IF.

Power, Sample size and MDE for CRTs based upon the Wald test from a linear mixed model

$\lambda = \frac{(\mu_1-\mu_2)^2}{2\frac{\sigma^2}{Km}[1+(m-1)\rho]}$ is a function of the design features K, m, σ^2 and ρ

- Power for a CRT based upon the Wald test for a linear mixed effects model is
  $\pi = 1 - Pr\{\chi^2[\lambda(K, m, \sigma^2, \rho, \beta^*), 1]\}$, where $\chi^2(\lambda, p)$ is the cumulative χ^2 distribution function with non-centrality parameter λ, and $p = 1$ degrees of freedom.
- To obtain the required number of clusters, K, for fixed cluster size m with σ^2 at the required power $\pi = 80\%$ and 5% significance level, we solve for K using $\lambda = 7.85 = \frac{(\beta^*)^2}{2\frac{\sigma^2}{Km}[1+(m-1)\rho]}$ to obtain
  $K = \frac{2\times 7.85\sigma^2[1+(m-1)\rho]}{m(\beta^*)^2}$, and $m = \frac{2\times 7.85\sigma^2(1-\rho)}{(\beta^{*2})K-2\times 7.85\sigma^2\rho}$.
- To find the MDE at the required power $\pi = 80\%$ at 5% significance level for fixed K,m, σ^2 and ρ, we set $\lambda = 7.85$ and obtain
  $MDE = \pm\sqrt{\frac{2\times 7.85\sigma^2[1+(m-1)\rho]}{Km}}$.

Power, Sample size and MDE for CRTs based upon the Wald test from a linear mixed model – adjustment for variable cluster size

- When it is expected that the cluster sizes will vary, substitute $\tilde{m}$ for m in the sample size formulas, where $\tilde{m} \approx \bar{m}(1 + CV_m^2)$. Variable cluster size reduces power, increases sample size requirements, and increases the $|MDE|$.
- λ now equals $\lambda = \frac{(\mu_1 - \mu_2)^2}{2\frac{\sigma^2}{K\bar{m}}[1+(\bar{m}(1+CV_m^2)-1)\rho]}$
- Thus, power is $\pi = 1 - Pr\{\chi^2[\lambda(K, \bar{m}, CV_m, \sigma^2, \rho, \beta^*), 1] \leq 3.84\}$
- For 80% power and 5% significance level, $K = \frac{2\times 7.85\sigma^2[1+(\bar{m}(1+CV_m^2)-1)\rho]}{\bar{m}(\beta^*)^2}$, and, solving for $\bar{m}$, $\bar{m} = \frac{2\times 7.85\sigma^2(1-\rho)}{K(\beta^*)^2 - 2\times 7.85\sigma^2(1+CV_m^2)\rho}$. Similarly, given K, $\bar{m}$, and CV_m if needed, the MDE, β^* can be solved for from either of these two equations.

Power, Sample size and MDE for CRTs based upon the Wald test from a linear mixed model

- In many applications, either m or K is already fixed by the study circumstances. For example, there may be only K clinics in the geographic region where the study is to take place.
- What if both m and K are free to be chosen by the investigator?
  - There is an entire surface of solutions (m, K) to the power formula in CRTs. To provide a unique (m, K) for a given setting, constrained optimization is used.
  - For example, we can solve for (m,K) as

$$\underset{m,K}{\arg\max}\{1 - Pr\{\chi^2[\lambda(K, m, \sigma^2, \rho, \beta^*), 1]\}\},$$

subject to the constraint that

$$Kc_1 + Kmc_2 \leq Cost,$$

where c_1 is the unit cost of adding a cluster, and c_2 is the unit cost of adding additional participants within a cluster.

Example of HPTN for minimizing cost subject to fixed power, π

- Recall, from the HPTN study, the binary response is having the risk injection behavior ($Y_{ijk} = 1$) or not ($Y_{ijk} = 0$) in the past 30 months.
- $X_{ijk} = 1$ for the participant in the intervention group, $X_{ijk} = 0$ for the participant in control group.
- To obtain possible parameter values to inform the design of a new study similar to HPTN, we fit GEE for binary data with the identity link, with Y_{ijk} as the dependent variable and X_{ijk} as the independent variable, where β^* is the RD.
- We obtained $\hat{\beta}^* = -0.0777$, $p_1 = \hat{\beta}_0 = 0.4885$ and $\hat{p}_2 = \hat{p}_1 + \hat{\beta}^* = 0.411$.
- Then, the average variance of a single observation is $\frac{\hat{p}_1(1-\hat{p}_1)+\hat{p}_2(1-\hat{p}_2)}{2} = 0.246$, and the ICC, $\hat{\rho} = 0.121$.

Example of HPTN for minimizing cost subject to fixed power

- We want to solve

$$\arg\min_{m,K}(Kc_1 + Kmc_2)$$

with $c_1 = \$10,000$ and $c_2 = \$50$, subject to constraints

$$\pi = 1 - Pr\{\chi^2[\lambda(K, m, \sigma^2, \rho, \beta^*), 1]\}$$

$$K > 0$$

$$m > 0$$

- Fixing $\pi = 80\%$, the constraint can be written as

$$\lambda = 7.85 = \frac{(\beta^*)^2}{2\frac{\sigma^2}{Km}[1 + (m-1)\rho]}$$

Example of HPTN for minimizing cost subject to fixed power

To solve this optimization problem, we can use the"optimize()" function in R

- Step 1: from the constraint we know:

$$K = \frac{2 \times 7.85 \times \sigma^2[1 + (m-1)\rho]}{(\beta^*)^2 \times m} \tag{2}$$

- Step 2: plug (2) into the objective function to eliminate K, we then have:

$$\arg\min_m \left\{ \frac{c_1 \times 2 \times 7.85 \times \sigma^2[1 + (m-1)\rho]}{(\beta^*)^2 \times m} + \frac{c_2 \times 2 \times 7.85 \times \sigma^2[1 + (m-1)\rho]}{(\beta^*)^2} \right\} \tag{3}$$

Example of HPTN for minimizing cost subject to fixed power

- Then the original optimization problem is changed to (3) with constraints:

$$K = \frac{2 \times 7.85 \times \sigma^2[1 + (m-1)\rho]}{(\beta^*)^2 \times m} > 0 \text{ and } m > 0$$

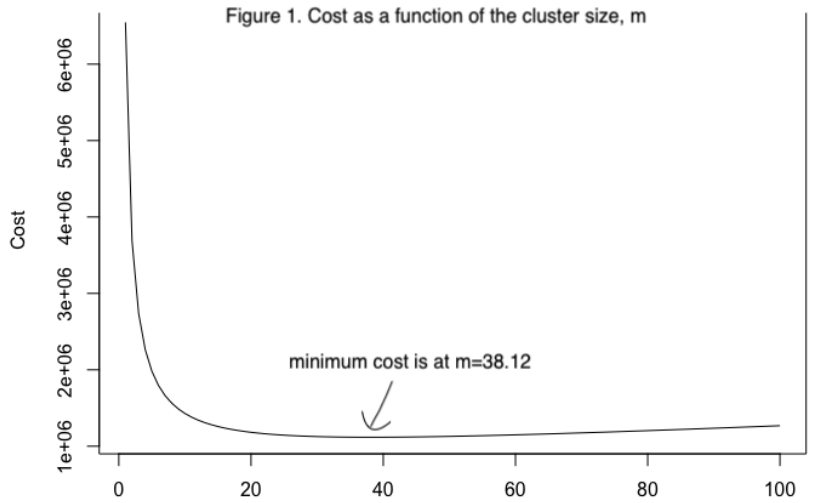
- Step 3: We first use optimize() to find m:

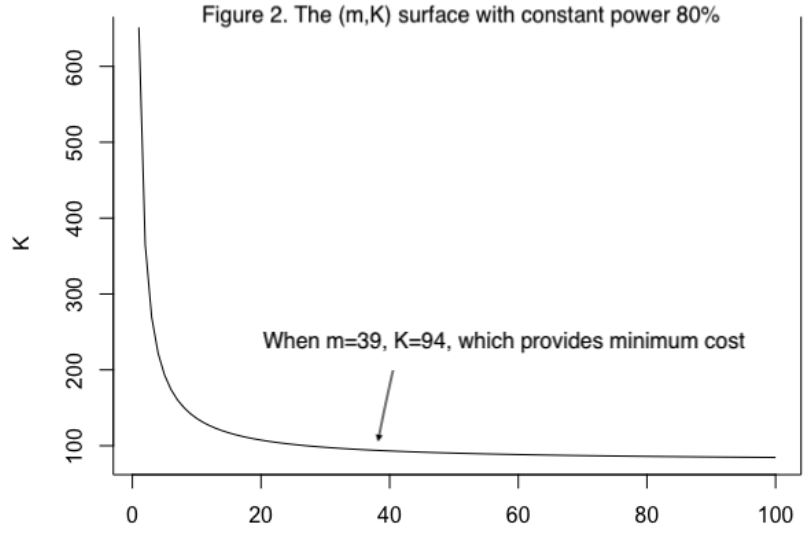
```
> C1=10000; C2=50
> f=function(m,A,B,r) {
+     objective=(15.7*C1*A*(1+(m-1)*r)+15.7*C2*A*(1+(m-1)*r)*m)/(B*m)
+     objective
+ }
>
> m = optimize(f,c(1,100),tol=0.0001,A=0.246,B=0.005929,r=0.121,lower=1)
> m
$minimum
[1] 38.11682

$objective
[1] 1117272
```

Example of HPTN for minimizing cost subject to fixed power

- After optimization, obtain $m = 38.12$. Using (2), we have $K = 93.84$, and the cost is \$1,117,272. $\lambda = 7.85$ and power $= 80\%$. Note: This m is unrealistic for the context. Probably should have constrained m to be ≤ 5.
- After rounding up to $K = 94$ and $m = 39$, cost is \$1,123,300. $\lambda = 7.891$, power $= 80.22\%$. If we round up to $K = 94$ and $m = 38$, we can obtain a lower cost \$1,118,600, $\lambda = 7.86$, and power $= 80.06\%$. If we round up to $K = 93$ and $m = 39$, we can obtain a lower cost \$1,111,350, but $\lambda = 7.81$ with power $= 79.8\%$.
- In HPTN 067, we actually have $K = 232$ and $\bar{m} = 3$, so the $\lambda = 6.75$, and the power is 73.9% for detecting a risk difference of $\hat{\beta}^* = -0.0777$. Given $C_1 = 10,000$ and $C_2 = 50$, $cost = \$2,320,150$.





Power, sample size and MDE for CRTs based upon the Wald test from a linear mixed model

- The cost criterion considers the cost of adding additional participants to clusters along with the costs of recruiting and maintaining new clusters. The latter is usually much more expensive, but the former has an asymptote as per Lecture notes 3.38-3.40
- Alternatively, sample size K for a CRT can be obtained by multiplying the per group sample size, n_1 for an RCT by the IF and dividing by m (in the absence of CRT software, for example). That is, per group sample size for a CRT is

$$n_1 \times IF = Km$$

- It can be shown that these results are obtained by asymptotic likelihood theory for cluster-correlated normally distributed $\boldsymbol{Y}$ from which the variance of the parameter of interest can be obtained.

Power, Sample size and MDE for CRTs with binary outcomes based upon a Wald test approximation from the linear mixed model

Now, $Y_{ijk} = 0$ or 1, a binary outcome. As always, $\lambda = \frac{(\beta^*)^2}{Var(\hat{\beta}^*)}$

- We take $\lambda = \frac{(\beta^*)^2}{Var(\hat{\beta}^*)} = \frac{(\beta^*)^2}{2\frac{\sigma^2}{Km}[1+(m-1)\rho]}$, and we substitute the variance for the difference between two binomial proportions from independent data for the variance for the difference between two means from independent data.
- To approximate the alternative variance under a comparison of two binomial proportions for independent data, we use $2\sigma^2 = p_1(1-p_1) + p_2(1-p_2)$, where $p_2 = \beta^* + p_1$, when β^* is the risk difference, and $p_2 = \beta^* p_1$, when β^* is the risk ratio.
- Then, $\lambda = \frac{(\beta^*)^2}{Var(\hat{\beta}^*)} = \frac{(\beta^*)^2}{\frac{p_1(1-p_1)+p_2(1-p_2)}{Km}[1+(m-1)\rho]}$

Power, Sample size and MDE for CRTs with binary outcomes based upon a Wald test approximation from the linear mixed model

- Use the same formulas as above for power, sample size and MDE, where now $\lambda = \frac{(\beta^*)^2}{Var(\hat{\beta}^*)} = \frac{(\beta^*)^2}{\frac{p_1(1-p_1)+p_2(1-p_2)}{Km}[1+(m-1)\rho]}$
- This is what PASS does
- This formula applies to linear mixed models for a binary outcome with the identity link (see Shih, 1997 for proof)

Brief sketch of GEE theory for clustered data with any type of outcome variable and link function

Let $E(\boldsymbol{Y}_{ik}|\boldsymbol{X}_{ik};\beta) = g^{-1}(\boldsymbol{X}_{ik};\beta)$. When $\boldsymbol{U}_{ik} \perp \boldsymbol{U}_{i'k'}\ \forall i \neq i'$ or $k \neq k'$, the solution, $\hat{\beta}$, to

$$\boldsymbol{U} = \Sigma_{i=1}^{2}\Sigma_{k=1}^{K}\boldsymbol{U_{ik}}(\boldsymbol{Y_{ik}}, \boldsymbol{X_{ik}}; \beta) = \Sigma_{i=1}^{2}\Sigma_{k=1}^{K}\boldsymbol{w_{ik}}(\boldsymbol{X}_{ik};\beta)^{T}(\boldsymbol{Y}_{ik} - E(\boldsymbol{Y}_{ik}|\boldsymbol{X}_{ik};\beta))) = \boldsymbol{0} \quad (4)$$

is consistent and asymptotically normal, where

$dim(\boldsymbol{Y}_{ik}) = dim(E(\boldsymbol{Y}_{ik}|\boldsymbol{X}_{ik})) = dim(\boldsymbol{g}(\boldsymbol{X}_{ik})) = m_{ik} \times 1, dim(\boldsymbol{X}_{ik}) = m_{ik} \times 2,$

$dim(\boldsymbol{w_{ik}}(\boldsymbol{X}_{ik};\beta)) = m_{ik} \times 2, dim(\boldsymbol{U}) = dim(\boldsymbol{U}_{ik}) = 2 \times 1$, and $\boldsymbol{Y}_{ik} - E(\boldsymbol{Y}_{ik}|\boldsymbol{X}_{ik};\beta) = \boldsymbol{e}_{ik}$ in a CRT.

It is easy to see that as long as $g^{-1}(\boldsymbol{X}_{ik};\beta)$ is correctly specified as the conditional mean of $\boldsymbol{Y}_{ik}|\boldsymbol{X}_{ik}\forall i, k$, then

$$E_{\boldsymbol{Y}|\boldsymbol{X}}(\boldsymbol{Y}_{ik} - E(\boldsymbol{Y}_{ik}|\boldsymbol{X}_{ik};\beta))) = \boldsymbol{0}.$$

Estimating functions that have this *unbiasedness* property have been shown under mild regularity conditions to produce consistent and asymptotically normal estimators (CAN).

Notes:

1. If $g(\boldsymbol{X}_{ik};\beta) = \boldsymbol{X}_{ik}^{T}\beta$, this is a generalized linear model
2. Most commonly used likelihood functions produce 'likelihood score' functions of form (4) above.
3. Even when the form (4) above does not hold, if the estimating equations have mean 0, consistent and asymptotically normal estimators will be obtained.
4. For scalar outcome, Y, we discussed this in Lecture 1, slide 33.

(Liang, K. Y. and S. L. Zeger (1986). "Longitudinal data-analysis using generalized linear models." Biometrika 73(1): 13-22.; Huber, P. (1967). The behavior of maximum likelihood estimates under nonstandard conditions. Fifth Berkeley Symposium on Mathematical Statistics and Probability, Berkeley, CA, University of California Press.)

Brief sketch of GEE theory for clustered data with any type of outcome variable and link function

- Often, $\boldsymbol{w_{ik}}(\boldsymbol{X}_{ik};\boldsymbol{\beta})^T = \boldsymbol{D}_{ik}\boldsymbol{\Sigma}_{ik}^{-1}$, where $\boldsymbol{D}_{ik} = \frac{\partial g^{-1}(\boldsymbol{X}_{ik};\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^T}$ and $\boldsymbol{\Sigma}_{ik} = diag^{1/2}[\sigma^2_{ijk}]\boldsymbol{R}_{ik}diag^{1/2}[\sigma^2_{ijk}]$.
- - $diag[\sigma^2_{ijk}]$ is a $m_{ik}xm_{ik}$ matrix with $Var(Y_{ijk})$ on the main diagonal and 0 otherwise
  - $\boldsymbol{R}_{ik}$ is a $m_{ik}xm_{ik}$ 'working correlation' matrix with 1 on the main diagonal and $Corr(Y_{ijk}, Y_{ij'k})$ on the off diagonals
  - These weights often maximize efficiency (e.g. MLE) but when correlation or variances are mis-specified estimating equations still provide consistent estimates
- $Var(\hat{\boldsymbol{\beta}})$ is estimated by

$$\widehat{Var}(\hat{\boldsymbol{\beta}}) \approx \Sigma_{i=1}^2\Sigma_{k=1}^K \left\{ \left[\frac{\partial \boldsymbol{U_{ik}}(\boldsymbol{Y_{ik}}, \boldsymbol{X_{ik}};\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right]^{-1} \boldsymbol{w_{ik}}(\boldsymbol{X}_{ik};\boldsymbol{\beta})^T \boldsymbol{e_{ik}}\boldsymbol{e_{ik}}^T \boldsymbol{w_{ik}}(\boldsymbol{X}_{ik};\boldsymbol{\beta}) \left[\frac{\partial \boldsymbol{U_{ik}}(\boldsymbol{Y_{ik}}, \boldsymbol{X_{ik}};\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right]^{-T} \right\} \Big|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}$$

- It can be difficult to obtain $Var(\hat{\boldsymbol{\beta}}) = E_{\boldsymbol{Y},\boldsymbol{X}}(\widehat{Var}(\hat{\boldsymbol{\beta}};\boldsymbol{Y},\boldsymbol{X}))$, where the complex expression for the estimator is given above
- Using this theory, it has been shown that when the correlation between participants in the same cluster $ik = \rho$ (and 0 otherwise of course), the results given above for the non-centrality parameter based upon the Wald test for the difference between two sample means given above are obtained (eq. 7 of Shih)!
- See Shih (1997)* for the derivations

*Shih, Weichung Joseph. "Sample size and power calculations for periodontal and other studies with clustered samples using the method of generalized estimating equations." Biometrical Journal 39.8 (1997): 899-908.

Derivation of the variance of the GEE estimator

The GEE estimator, $\hat{\boldsymbol{\beta}}$, solves

$$\boldsymbol{U}(\boldsymbol{\beta}) = \sum_{i=1}^{2}\sum_{k=1}^{K} \boldsymbol{U}_{ik}(\boldsymbol{\beta}) = \sum_{i=1}^{2}\sum_{k=1}^{K} \boldsymbol{D}_{ik}^{T}(\boldsymbol{\beta})\, \boldsymbol{\Sigma}_{ik}^{-1} \Big(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}) \Big) = \boldsymbol{0},$$

where $\boldsymbol{\mu}_{ik}(\boldsymbol{\beta}) = E(\boldsymbol{Y}_{ik} \mid \boldsymbol{X}_{ik}; \boldsymbol{\beta})$, $\boldsymbol{D}_{ik}(\boldsymbol{\beta}) = \dfrac{\partial \boldsymbol{\mu}_{ik}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^{\top}}$, and $\boldsymbol{w}_{ik}(\boldsymbol{\beta})$, defined on the previous slide, is $\boldsymbol{D}_{ik}^{T}(\boldsymbol{\beta})\, \boldsymbol{\Sigma}_{ik}^{-1}$.

With a first order Taylor series expansion of $\boldsymbol{U}(\hat{\boldsymbol{\beta}})$ around $\boldsymbol{\beta}_0$, we have that

$$\boldsymbol{0} = \boldsymbol{U}(\hat{\boldsymbol{\beta}}) \approx \boldsymbol{U}(\boldsymbol{\beta}_0) + \frac{\partial \boldsymbol{U}(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}} \Big(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0 \Big).$$

and

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0 \approx - \left[\frac{\partial \boldsymbol{U}(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}} \right]^{-1} \boldsymbol{U}(\boldsymbol{\beta}_0),$$

$$\sqrt{2K}\,(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \approx - \left[\frac{1}{2K} \sum_{i=1}^{2}\sum_{k=1}^{K} \frac{\partial \boldsymbol{U}_{ik}(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}} \right]^{-1} \left(\frac{1}{\sqrt{2K}}\, \boldsymbol{U}(\boldsymbol{\beta}_0) \right).$$

Derivation of the variance of the GEE estimator

For each (i, k),

$$\frac{\partial \boldsymbol{U}_{ik}(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}} = \frac{\partial}{\partial \boldsymbol{\beta}} \left\{ \boldsymbol{D}_{ik}^T(\boldsymbol{\beta})\, \boldsymbol{\Sigma}_{ik}^{-1} \Big(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}) \Big) \right\} \Big|_{\boldsymbol{\beta}=\boldsymbol{\beta}_0}$$

$$= \underbrace{\frac{\partial \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}}\, \boldsymbol{\Sigma}_{ik}^{-1} \Big(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0) \Big)}_{\text{Term A}} - \underbrace{\boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\, \boldsymbol{\Sigma}_{ik}^{-1}\, \frac{\partial \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}}}_{\text{Term B}}.$$

Since $\dfrac{\partial \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}} = \boldsymbol{D}_{ik}(\boldsymbol{\beta}_0)$, Term B equals

$$\text{Term B} = \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\, \boldsymbol{\Sigma}_{ik}^{-1}\, \boldsymbol{D}_{ik}(\boldsymbol{\beta}_0).$$

For Term A, average over (i, k). By definition, $E\Big[\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0) \Big] = \boldsymbol{0}$:

$$\frac{1}{2K} \sum_{i=1}^{2} \sum_{k=1}^{K} \text{Term A} \xrightarrow{p} E\left[\frac{\partial \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}}\, \boldsymbol{\Sigma}_{ik}^{-1} \Big(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0) \Big) \right] = \boldsymbol{0}.$$

Therefore,

$$-\frac{1}{2K} \sum_{i=1}^{2} \sum_{k=1}^{K} \frac{\partial \boldsymbol{U}_{ik}(\boldsymbol{\beta}_0)}{\partial \boldsymbol{\beta}} = -\frac{1}{2K} \sum_{i=1}^{2} \sum_{k=1}^{K} \Big(\text{Term A} - \text{Term B} \Big)$$

$$\approx \frac{1}{2K} \sum_{i=1}^{2} \sum_{k=1}^{K} \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\, \boldsymbol{\Sigma}_{ik}^{-1}\, \boldsymbol{D}_{ik}(\boldsymbol{\beta}_0).$$

Derivation of the variance of the GEE estimator

Then

$$\sqrt{2K}\,(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \approx \left[\frac{1}{2K}\sum_{i=1}^{2}\sum_{k=1}^{K} \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\,\boldsymbol{D}_{ik}(\boldsymbol{\beta}_0)\right]^{-1} \left(\frac{1}{\sqrt{2K}}\,\boldsymbol{U}(\boldsymbol{\beta}_0)\right)$$

So

$$\begin{aligned}
&\mathrm{Cov}(\sqrt{2K}\,(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)) \\
&\approx \left[\frac{1}{2K}\sum_{i=1}^{2}\sum_{k=1}^{K} \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\,\boldsymbol{D}_{ik}(\boldsymbol{\beta}_0)\right]^{-1} \mathrm{Cov}\left(\frac{1}{\sqrt{2K}}\,\boldsymbol{U}(\boldsymbol{\beta}_0)\right) \left[\frac{1}{2K}\sum_{i=1}^{2}\sum_{k=1}^{K} \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\,\boldsymbol{D}_{ik}(\boldsymbol{\beta}_0)\right]^{-1}.
\end{aligned}$$

According to

$$\begin{aligned}
\mathrm{Cov}\left(\frac{1}{\sqrt{2K}}\,\boldsymbol{U}(\boldsymbol{\beta}_0)\right) &= \mathrm{Cov}\left(\frac{1}{\sqrt{2K}}\sum_{i=1}^{2}\sum_{k=1}^{K} \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\left(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0)\right)\right) \\
&= \frac{1}{2K}\sum_{i=1}^{2}\sum_{k=1}^{K} \mathrm{Cov}\left(\boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\left(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0)\right)\right) \\
&= \frac{1}{2K}\sum_{i=1}^{2}\sum_{k=1}^{K} \boldsymbol{D}_{ik}^T(\boldsymbol{\beta}_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\,\mathrm{Cov}\left(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\boldsymbol{\beta}_0)\right)\boldsymbol{\Sigma}_{ik}^{-1}\boldsymbol{D}_{ik}(\boldsymbol{\beta}_0).
\end{aligned}$$

Derivation of the variance of the GEE estimator

Since $\text{Cov}(\boldsymbol{Y}_{ik} - \boldsymbol{\mu}_{ik}(\beta_0)) = \text{Cov}(\boldsymbol{Y}_{ik}) = \boldsymbol{\Sigma}_{ik}$,

$$\text{Cov}\left(\frac{1}{\sqrt{2K}}\boldsymbol{U}(\beta_0)\right) = \frac{1}{2K}\sum_{i=1}^{2}\sum_{k=1}^{K}\boldsymbol{D}_{ik}^{T}(\beta_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\boldsymbol{D}_{ik}(\beta_0)$$

Then we have

$$\text{Cov}\left(\sqrt{2K}\,(\hat{\beta}-\beta_0)\right) \approx \left[\frac{1}{2K}\sum_{i=1}^{2}\sum_{k=1}^{K}\boldsymbol{D}_{ik}^{T}(\beta_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\,\boldsymbol{D}_{ik}(\beta_0)\right]^{-1}.$$

so that

$$\text{Cov}(\hat{\beta}) \approx \left[\sum_{i=1}^{2}\sum_{k=1}^{K}\boldsymbol{D}_{ik}^{T}(\beta_0)\,\boldsymbol{\Sigma}_{ik}^{-1}\,\boldsymbol{D}_{ik}(\beta_0)\right]^{-1}.$$

GEE theory to obtain the non-centrality parameter for binary data

Similarly, for binary data, the logistic link is typically used, thereby estimating the odds ratio, rather than the identity link, thereby estimating the risk difference. When the logistic link is assumed, i.e. if $E(Y_{ijk}|X_{ijk}) = g^{-1}(X_{ijk};\beta) = \frac{exp^{\beta_0+\beta^* X_{ijk}}}{1+exp^{\beta_0+\beta^* X_{ijk}}}$, Shih (1997) shows using the above theory that the non-centrality parameter based upon the Wald test for $H_0 : OR = exp^{\beta*} = 1$ vs $H_A : OR = exp^{\beta*} \neq 1$ is obtained as

$$\lambda = \frac{(log(OR))^2}{Var(log(\widehat{OR}))} = \frac{(log(OR))^2}{\frac{[p_1(1-p_1)]^{-1}+[p_2(1-p_2)]^{-1}}{Km}[1+(m-1)\rho]}$$

instead of

$$\lambda = \frac{(\beta^*)^2}{Var(\hat{\beta}^*)} = \frac{(\beta^*)^2}{\frac{p_1(1-p_1)+p_2(1-p_2)}{Km}[1+(m-1)\rho]},$$

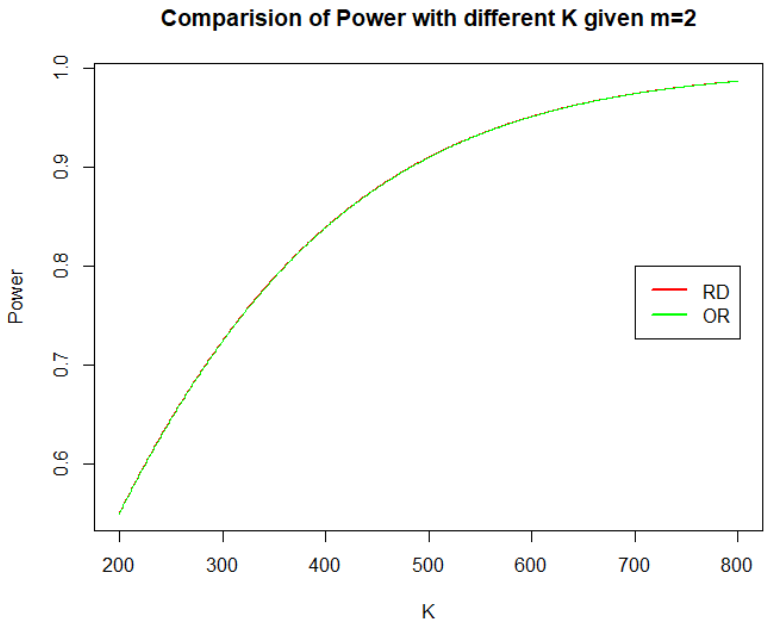
where $Var(log(\widehat{OR})) = [p_1(1-p_1)]^{-1} + [p_2(1-p_2)]^{-1}$ is the Woolf's variance for the log odds ratio estimator

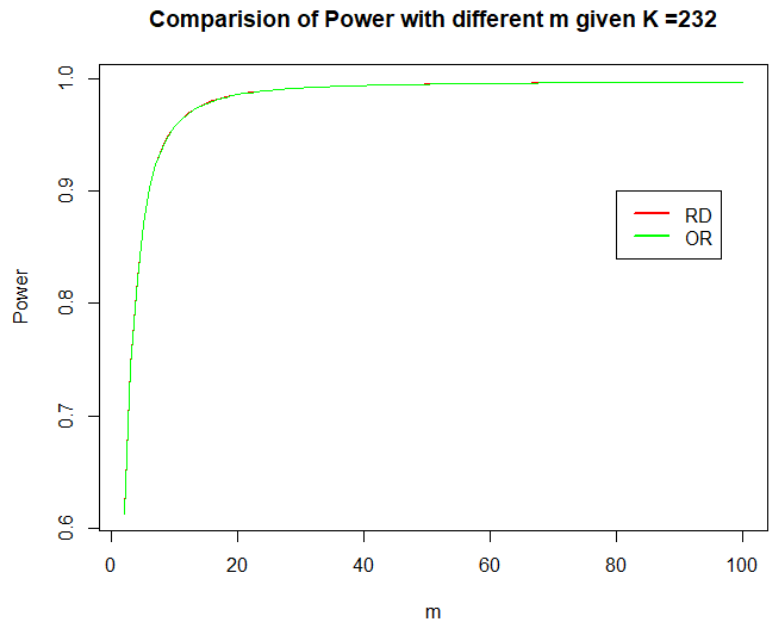
Efficiency implications of the choice of link function

- It is easy to show that basing the construction of the asymptotic Wald test statistic will lead to a reduction in efficiency compared to the test based on the identity link
- Since $1/p_1 > 1$, $1/(1-p_1) > 1$, $p_1 < 1$, $(1-p_1) < 1$, $1/p_1 * 1/(1-p_1) > 1 > p_1(1-p_1) < 1$, and likewise for p_2.
- Hence $Var(log(\widehat{OR})) = [p_1(1-p_1)]^{-1} + [p_2(1-p_2)]^{-1} > p_1(1-p_1) + p_2(1-p_2) = Var(\hat{p}_2 - \hat{p}_1) = Var(\widehat{RD})$
- The denominators of the two Wald test statistics are based upon the variance of the numerator, here the variance of the $log(\widehat{OR})$ and $\widehat{RD}$), respectively.

Comparison of the power between the logit vs identity link for binary data

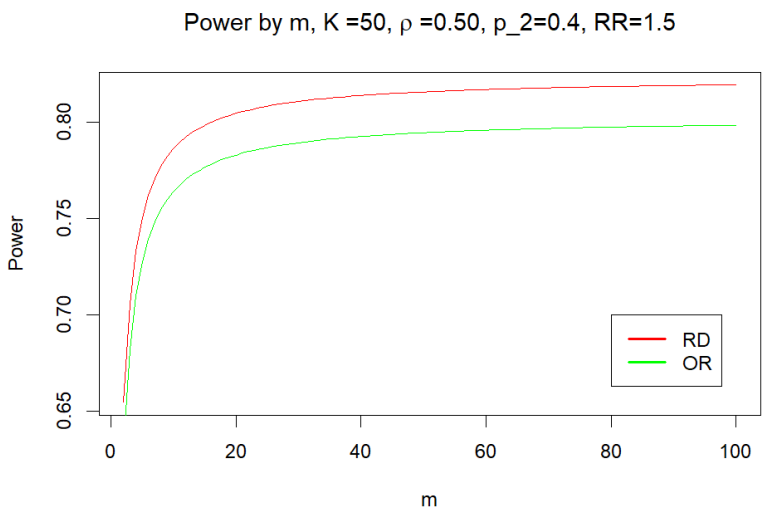
- Now we use the HPTN input parameters to compare the power between the logit and identity links for binary data
- Recall in HPTN data: $\bar{m} = 2$, $K = 232$ and $\rho = 0.121$
- Using identity link, we have $\widehat{\beta}^* = -0.0775$, $p_1 = 0.4885$, $p_2 = 0.411$
- Using logit link, we have $log(\widehat{OR}) = -0.3148$, $p_1 = 0.488$, $p_2 = 0.411$

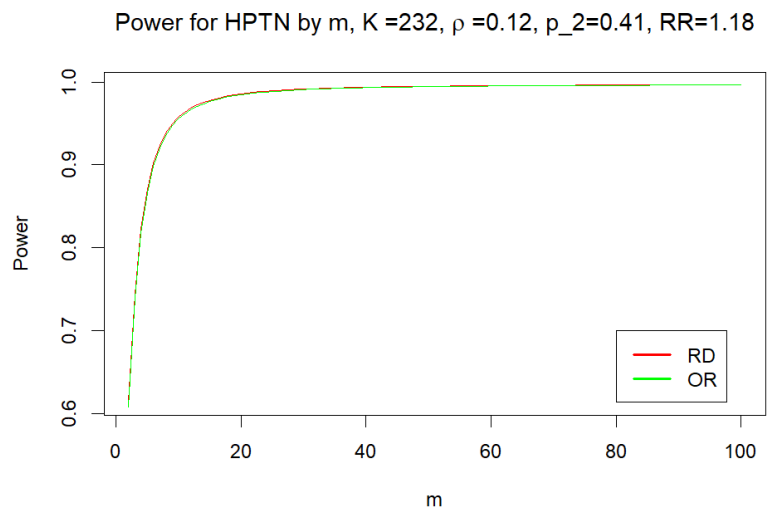




Comparison of the power between the logit vs identity link for binary data

- Continuing our comparison of the power between tests arising from the logit and identity links for binary data, we compare the HPTN input parameters to another lower power case
- Recall in HPTN, $\bar{m} = 2$, $K = 232$ and $\rho = 0.121$. Now, we change to K=50, $\rho = 0.5$
- Using the identity link, we estimated in HPTN that $p_1 = 0.4885$, $p_2 = 0.411$. Now, we change to $p_1 = 0.6$ and p_2=0.4





Conditional vs. marginal models

- $E(Y_{ijk}|\boldsymbol{X}_{ijk};\beta) = g^{-1}(\boldsymbol{X}_{ijk};\beta)$ is a conditional mean model of marginal form. Marginal is over the distribution of the random effect, $\boldsymbol{b_{ik}}, f(\boldsymbol{b_{ik}})$, where $\boldsymbol{b}_{ik} = (b_{ik}, ..., b_{ik})^T$ with dimension $m_{ik} \times 1$
  - GEE are marginal models
  - These are also called cluster-specific models and population-averaged models.
- $E(Y_{ijk}|\boldsymbol{X}_{ijk}, b_{ik};\beta) = g^{-1}(\boldsymbol{X}_{ijk}, b_{ik};\beta)$ is a conditional mean model of conditional form. Conditional is conditional on the random effect, b_{ik}. This means that the assumed structure for the dependency of the mean of the outcome on the covariates has form $g^{-1}(\cdot)$ within its cluster.
  - These are also called subject-specific models
- When are these models 'the same'? By 'the same', we might mean, when does $E_b(\boldsymbol{Y}_{ijk}|\boldsymbol{X}_{jik}, b_{ik};\beta) = g^{\#(-1)}(\boldsymbol{X}_{ijk};\beta^{\#}) = \int_b g^{-1}(\boldsymbol{X}_{ijk}, b;\beta) f(\boldsymbol{b_{ik}}) db = g^{-1}(\boldsymbol{X}_{ijk};\beta)$
- Or, perhaps less restrictively, we may ask when does $\beta = \beta^{\#}$?
- Or, even less restrictively, when does $\beta^* = \beta^{\#^*}$?

Conditional vs. marginal models

- Recall the linear mixed model $Y_{ijk} = \beta_0 + \beta^* X_{ijk} + b_{ik} + e_{ijk}$.
- Then the subject-specific model is
  $E_e[Y_{ijk}|X_{ijk}, b_{ik}] = E_e[\beta_0 + \beta^* X_{ijk} + b_{ik} + e_{ijk}] = \beta_0 + \beta^* X_{ijk} + b_{ik}$,
  where it is assumed, as usual, that $Corr(e_{ijk}, b_{ik}) = 0$ for all i, j, k.
- $E_{b,e}[Y_{ijk}|X_{ijk}, b_{ik}, e_{ijk}] = E_{b,e}[\beta_0 + \beta^* X_{ijk} + b_{ik} + e_{ijk}] =$
  $E_b[\beta_0 + \beta^* X_{ijk} + b_{ik}] = \beta_0 + \beta^* X_{ijk}$. This is the population-averaged model.
- Since the conditional and marginal models have the same form with respect to the fixed effects, with a constant intervention effect multiplied by the intervention, these models estimate the same fixed effect parameters.
  - That is, with the identity link function, the marginal and conditional models estimate the same intervention effect.
  - These results are distribution free assuming $E(b_{ik}) = 0$ and $Var(b_{ik}) = \sigma^2_B$ and apply to any dependent variable, continuous, binary or categorical

Conditional vs. marginal models

Now, let's consider the subject-specific (aka conditional as opposed to marginal) logistic regression model,

$$E(Y_{ijk}|X_{ijk}, b_{ik}; \beta^{\#}) = \frac{exp(\beta_0^{\#} + \beta_1^{\#} X_{ijk} + b_{ik})}{1 + exp(\beta_0^{\#} + \beta_1^{\#} X_{ijk} + b_{ik})}.$$

To obtain the population-averaged model, we need to average over the random effect, b_{ik}

$$E(Y_{ijk}|X_{ijk}; \beta^{\#}) = E_b(Y_{ijk}|X_{ijk}, b_{ik}; \beta^{\#}) =$$
$$\int_b \frac{exp(\beta_0^{\#} + \beta_1^{\#} X_{ijk} + b_{ik})}{1 + exp(\beta_0^{\#} + \beta_1^{\#} X_{ijk} + b_{ik})} f(b; \sigma_B^2) db$$

If $f(b; \sigma_B^2)$ is $N(0, \sigma_B^2)$ as is often assumed, there is no closed form solution to this expression.

- In fact, under these circumstances, it has been shown that $\beta^{\#*} > \beta^*$
- When $\beta^{\#*} \neq \beta^*$, the population-averaged estimate may be difficult to interpret, as it is unclear what population is being averaged over and when this, whatever it is, is a target population of interest.

Conditional vs. marginal models

These results have implications for the design of cluster-randomized and stepped-wedge trials of binary outcomes:

- With binary outcomes and a normal random effects assumption, optimal designs (power, sample size, minimum detectable effect size) will be different when linearity with the logistic link function is assumed conditionally or marginally on the random effect
- All of our formulas up to now for design of CRTs with binary outcomes apply to the marginal model with the identity link. The extent to which design results would be different under a logistic link function is unknown. (Could be a nice research project!)
- The subject-specific model has a direct causal interpretation in terms of counterfactual outcomes.
- The results will be the same under the linear link function for binary data (e.g. when the risk difference is the parameter of interest; see slide 12)
- Ritz and Spiegelman (2004) show that the slope parameter (log relative risk) has the same interpretation for the log link function.

Citation: Ritz, J. and D. Spiegelman (2004). "Equivalence of conditional and marginal regression models for clustered and longitudinal data." Statistical Methods in Medical Research 13(4): 309-323.